

Chandan Kumar Singh

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Summary

Software Engineer with 1+ year experience designing scalable distributed systems using Java, Spring Boot, Kafka, AWS; delivered 40-50% latency reduction and 60% overhead cut through high-ownership engineering. Built production-grade microservices and AI-powered platforms. Strong foundation in Python, C++, and data streaming; 1000+ DSA solved | CodeChef Global Rank 14.

Experience

Tata Consultancy Services

Software Engineer — Java, Spring Boot, Microservices

Jan 2025 – Present

- Scalable API Design: Designed and owned RESTful APIs serving business-critical banking workflows; reduced average response latency by 40% and improved platform scalability by 50% through service-layer optimizations.
- API Gateway & System Design: Architected a centralized API Gateway with secret-key authentication and JWT token-reuse strategy, cutting inter-service overhead by 60% and improving distributed system reliability.
- Security Engineering: Engineered secure integrations with external banking systems using AES-256 encryption and defensive patterns against MITM, Replay, and Request Tampering attacks, ensuring zero security incidents.
- Automation & Ownership: Owned end-to-end delivery of internal automation solutions, eliminating 64% of manual operational effort and improving team release velocity.
- High-Throughput Engineering: Engineered a Bulk Upload pipeline for concurrent ingestion of thousands of records per request, improving system throughput by 35% with zero data-consistency issues.

Projects

Clothify Shopping Application | Java, Spring Boot, Hibernate, PostgreSQL, React.js, gRPC, Kafka, Docker, Kubernetes, AWS

- Distributed Microservices Design: Built a scalable e-commerce backend with API Gateway routing, Eureka service discovery, and LLD-driven domain separation across product, user, and order services.
- Event-Driven & Cloud-Native Deployment: Engineered low-latency gRPC + Kafka-based async communication with JWT security; containerized and deployed on Kubernetes (AWS EC2) via Jenkins CI/CD, sustaining 100+ concurrent users with 30% latency reduction.

QuantPulse AI | Python, Java, Spring Boot, FastAPI, React.js, XGBoost, Random Forest, FinBERT, PostgreSQL, Redis, WebSockets

- Ensemble ML Engine: Built a stock prediction engine combining XGBoost, Random Forest, and Logistic Regression on 60+ technical indicators with FinBERT-based news sentiment, improving prediction accuracy over single-model baselines.
- Real-Time Streaming Architecture: Architected Spring Boot + FastAPI + React with WebSocket-based tick-by-tick market data streaming, Redis caching, and resilient Groww API integration using exponential backoff.

Technical Skills

Languages: Python, C++, Java, JavaScript, TypeScript, SQL

Frameworks: Spring Boot, Spring Security, Hibernate, FastAPI, gRPC, Kafka, REST API, React.js

Databases: PostgreSQL, MySQL, Redis, FAISS, ChromaDB

Cloud: AWS (EC2, S3), Docker, Kubernetes, Jenkins

Tools: Git, GitHub, Docker, Kubernetes, Jenkins, NumPy, Pandas, Scikit-learn, HuggingFace, Transformers, LangChain, SVN

Education

Motihari College of Engineering, Motihari

B.Tech in Computer Science and Engineering (CSE)

Oct 2020 – Aug 2024